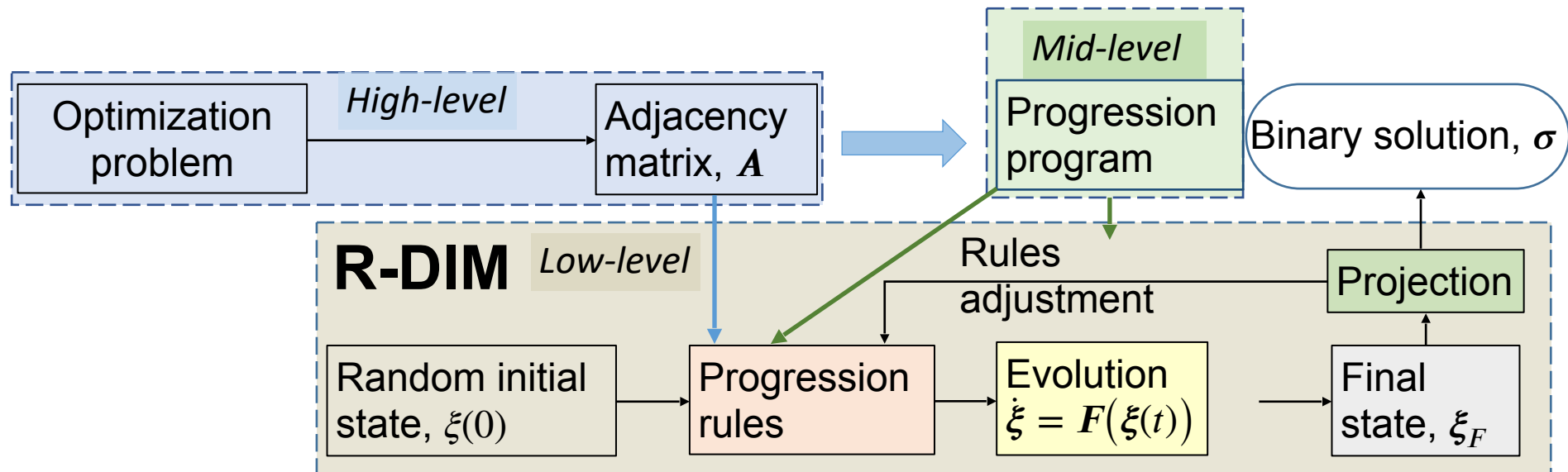


MCU-platform for R-DIM

University of Michigan

R-DIM: Relaxation-based DIM



Software-hardware co-development

Hardware component

Core architecture

Core instruction set
Heterogeneous kernels
Adaptive control

Multi-board architectures

Extended instruction set
Communication protocols
Fixed and expandable architectures

Application-centered architectures

Specialized instruction sets
Optimized nodes layouts
Hyperparallelization techniques

Software component

User-facing interfaces

C/C++, Python, Julia bindings, ...

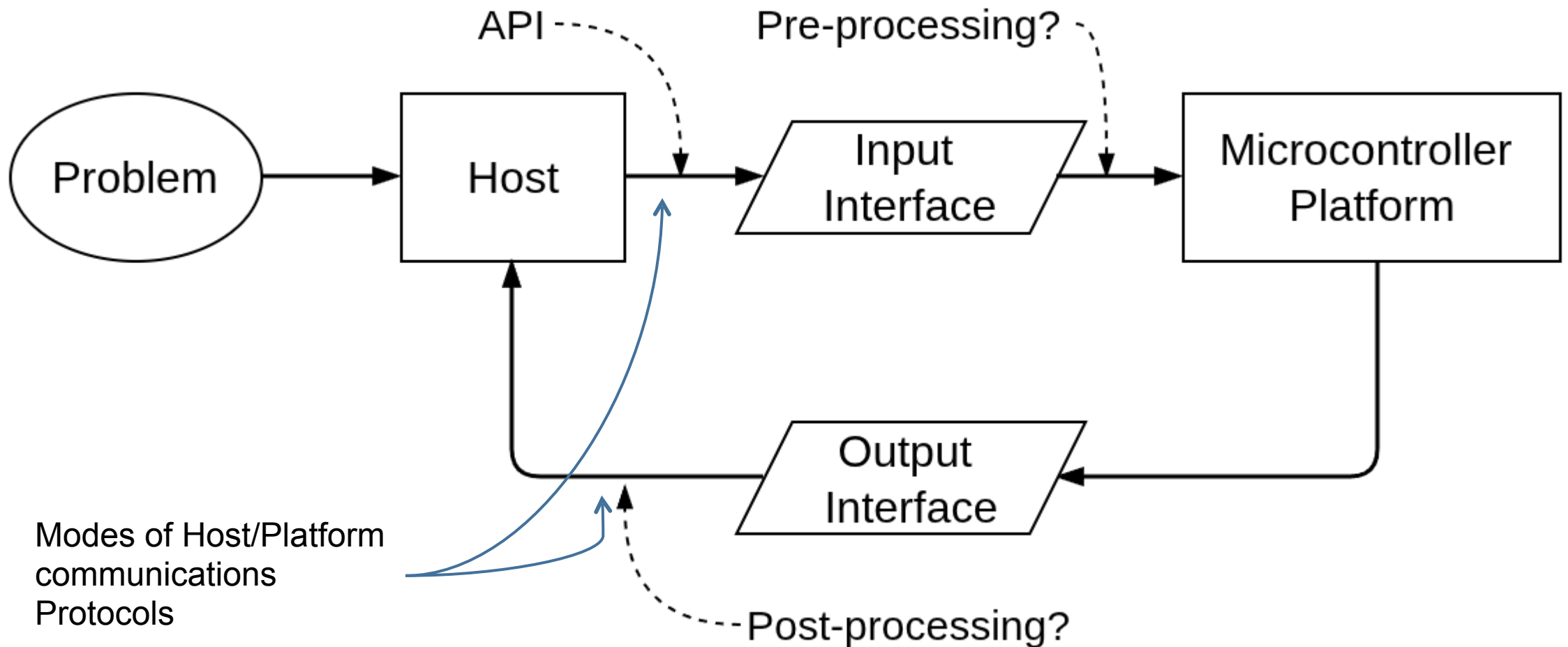
Pre-processing tools

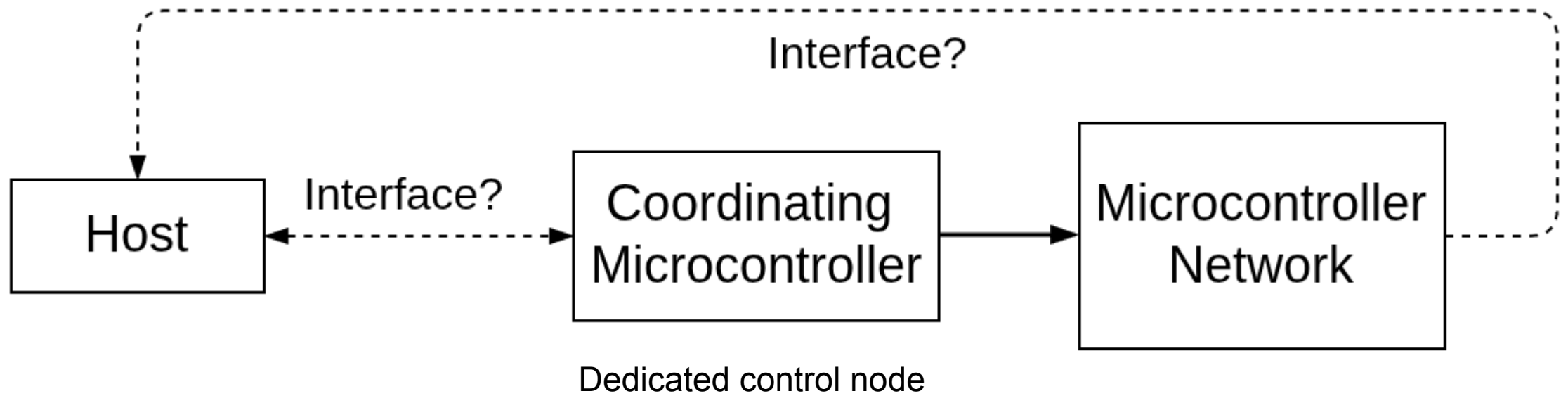
Mid-level and low-level code generators, ...

Behavior simulation platform

Platform emulator, ...

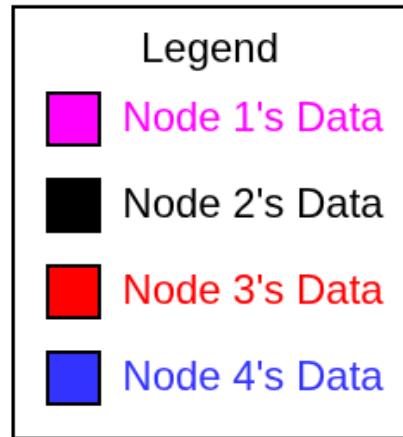
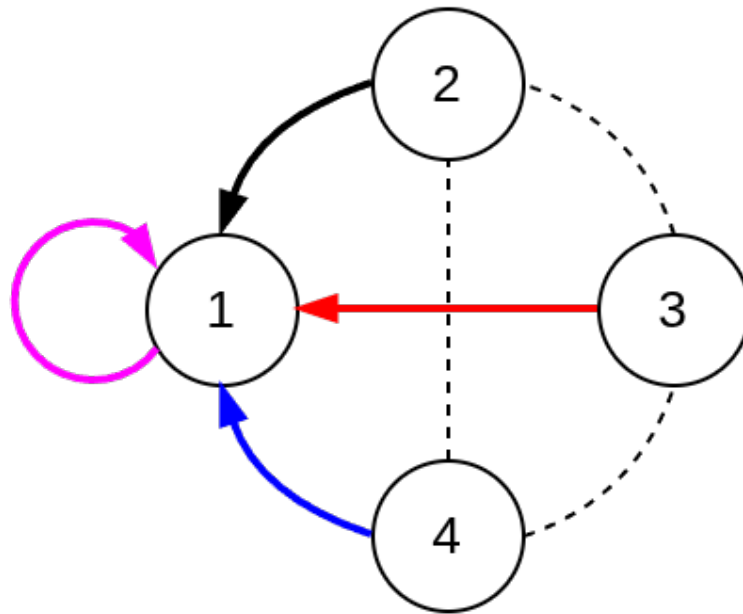
Platform dataflow overview





Topologies: network vs platform

Network topology and the order of processing



Platform topology and data pass-through

